KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 1st Term Examination, 2023
Math 3113
(Statistics and Numerical Methods)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). In an industrial area, there are two heavy welding factories, W_1 and W_2 . The average yearly repaired product and standard deviations for each factory are as follows : 10

Factory	Average production (In thousand)	Standard deviation (In Thousand)	No. of welding machine
W_1	20	3	200
W_2	30	2	80

(i) In which factory is there greater variation in the distribution of daily production of machine?
(ii) What are the mean and standard deviation (combined) of all the production in the industry? (Using formula of combined mean and variance).

1(b). In a research-based energy production plant, a scientist records the highest level of energy in each 1-minute interval. The observations are given below: 25

Range of energy level	2 − 4	5 − 6	7 − 9
Numbers of times observed	3	5	3

2(a). Define the standard deviation and variance. The delay times (Handling, setting, and poisitioning the tools) for cutting 6 parts on an engine lathe are 0.6, 1.2, 0.9, 1.0, 0.6 and 0.8 minutes. Calculate the variance and the standard deviation. 15

2(b). Determine the first, second, third and fourth moments about the origin 4 for the set of numbers 2, 3, 7, 8, 10. 10

2(c). A server has three data supply units-A, B, and C- from which it receives 60%, 30%, and 10% of the data, respectively. It has been observed that 15% of the data from A, 20% form B and 5% form C are defective. Given that data is received by the server, calculate the probability that the data is both defective and originates form unit A. Also, determine the overall probability of the data received by the server being defective. 10

3(a). TMR has two steel factories A and B with average weekly wages and standard deviations as follows: 15

Factory	Average wages	Standard deviation	No. of workers
T A	10	5	50
T B	30	4	5

Suppose in factory TB, the daily wages of a worker was wrongly posted noted as Tk 15 instead of Tk 55. What are the current mean and variance for factory B? Also, find the combind mean(i) before correction and (ii) after correction.

3(b). Define the conditional probability. If the probability that a communication will have high fidelity is 0.81 and the probability that it will have high fidelity and high selectivity is 0.18, what is the probability that a system with high fidelity will also have high selectivity? 05

3(c). The daily rainfall (in inches) in a certain region is continuous random variable x with probability density function given by 15

$$f(x) = \begin{cases} \frac{3}{4}(2x - x^2); & \text{if } 0 < x < 2 \\ 0 \quad ; & \text{otherwise} \end{cases}$$

4(a). Test whether the following function is probability function (pmf) or not, 10

x	-1	0	1	2
p(x)	2k	0.1	k	0.4

Hence, if possible, find i). the mean standard deviation and the moment coefficint of kurtosis.

4(b). An observer counted the number of defective items of a production machine in each 30-minute duration which is shown in the following table 10

The number of defective items	0	1	2	3
Observed in each 30-minute time span	2	5	4	2

i). Find the probability that two defective items observed within 2 hours.

ii). Find the probability that the first defective item observed within the duration 10.00 to 12.00 a.m

iii). Find the probability that no any defective item within the duration 10.00 to 12.00 a.m

iv). Find the probability that two defective items observed within the duration 10.00 to 12.00 a.m, and within the duration 1.00 to 2.00 p.m .

4(c). The grade on a class test were 0, 1, 2..., 10. The mean grade was 7.4 and the standard deviation was 1.5. Assume that the grade is normally distributed. Determine, 08

(i) the probability of students scoring 8, and

(ii) the maximum grade of the lowest 20% of the class. Also, find the estimated number of students scoring 7 among 30 students. (Using Z table)

4(d). In an automatic traffic control system, car speeds are monitored for over-speeded. Based on previous analysis, the probability of a car being over-speeded is 0.30. In a specific time frame, four cars are investigated. Calculate the probabilities for the following scenarios: 07

i). Green light (no over speeded cars observed).

ii). Red light (all four cars are over speeded).

iii). Otherwise (mixed scenarios of over speeded and non-over speeded cars).

SECTION – B

5(a). Discuss the different types of errors in numerical calculation. Mention some numerical techniques for finding approximate roots of nonlinear equation. 10

5(b). Find a root of $x \sin x - 1 = 0$ in the interval $[0, 2]$ by using the bisection method upto five steps. 10

5(c). Obtain the real root of the equation $x^2 + 4 \sin x = 0$ correct to four places of decimals by using Newton-Raphson method starting with $x_0 = -1.9$. Why do you need to choose the negative value of x? 15

6(a). Test whether the following system of linear equations is stricly diagonally dominant or not and hence solve the system of equations by the Gauss-Seidel method: 12

$$5x_1 + x_2 + 2x_3 = 19$$
$$x_1 + 4x_2 - 2x_3 = -2$$
$$2x_1 + 3x_2 + 8x_3 = 39$$

6(b). Determine the roots of the system of nonlinear equations 13

$$x^2 + y^2 = 5$$
$$x^2 - y^2 = 1$$

Using Newton-Rapshon method assuming $x_0 = 1$ and $y_0 = 1$

6(c). The values of x and y are given as follows 10

x :	5	6	9	11
y :	12	13	14	16

Find the value of y when x = 10, using Lagrange's interpolation formula.

7(a). Evaluate the first and second derivatives of the function $f(x)$ tabulated below at the point $x = 1.1$: 10

x	1	1.2	1.4	1.6	1.8	2.0
$f(x)$	0	0.1280	0.5440	1.2960	2.4320	4.0000

7(b). The temperature of a metal strip was measured at various time intervals during heating and the values are given in the following table: 13

Time t (min)	1	2	3	4
Temperature T (°C)	70	82	100	120

If the relationship between the temperature T and time t is of the form $T = be^{t/4} + a$, estimate the temperature at $t = 10$ min.

7(c). Solve the following system of linear equations by using LU decomposition method. 12

$$2x + y + z = 5$$
$$3x + 5y + 2z = 15$$
$$2x + y + 4z = 8$$

8(a). Evaluate $\int_0^1 \frac{\ln(1+x^2)}{1+x^2}dx$ by using Simpson's $\frac{3}{8}$ rule dividing the whole range into ten equal parts upto four decimal places. 10

8(b). The following table gives the population of a town during the last six censures. Estimate using any suitable interpolation formula, the increase in the population during the period from 1946 to 1948: 10

Year	1911	1921	1931	1941	1951	1961
Population (in thousand)	12	15	20	27	39	52

8(c). Use the fourth-order Runge-Kutta method to estimate $y(0.4)$ when 07

$$\frac{dy}{dx} = x^2 + y^2, y(0) = 0 \text{ and } h = 0.2$$

8(d). Use Taylor series method to solve the equation 08

$\frac{dy}{dx} = x + y, \quad y(1) = 0$ numerically upto $x = 1.2$ with $h = 0.1$

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2023

ESE 3107

(Solar Thermal Engineering)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks. Required graph /table will be provided
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Define the following terms: 10

(i) Solar constant, (ii) Azimuth angle, (iii) Irradiation, and (iv) Air mass.

1(b). Calculate the angle of incidence of beam radiation on a surface located at Nirala (22.8°N) at 11:00 (solar time) on February 15, if the surface is tilled 25° from the horizontal and pointed 8° west of south. Also calculate the geometric factor for this surface. Given that, 12

$$\cos\theta = -\sin\delta \cos\varphi \cos\gamma + \cos\delta \sin\varphi \cos\gamma \cos\omega + \cos\delta \sin\gamma \sin\omega$$

All the symbols represent typical notations.

1(c). What is sunshine hour? How it is related to declination? Explain. 06

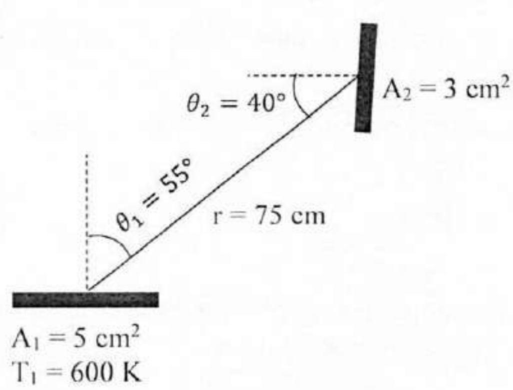
1(d). Differentiate between terrestrial and extraterrestrial radiation. 07

2(a). Define: (i) Beam radiation, (ii) Global radiation and (iii) Diffuse fraction. 07

How is daily diffuse fraction related with daily clearness index? Explain.

2(b). What is blackbody? Show that the emissive power of a blackbody is π times of radiation intensity. 08

2(c). A small surface of area A_1 = 5 cm^2 emits radiation as a blackbody at 600 K. Part of the radiation emitted by A_1 strikes another small surface of area A_2 = 3 cm^2 oriented as shown in the figure. Determine: 12
(i) the spectral blackbody emissive power of surface A_1 at a wavelength of 3 μm,
(ii) solid angle subtended by A_2 when viewed from A_1,
(iii) the rate at which radiation emitted by A_1 strikes A_2.



2(d). Briefly describe with necessary equations: 08

(i) Monthly average hourly clearness index

(ii) Overall tilt factor.

3(a). Define view factor. Explain crossed-strings method of finding view factor. 05

3(b). Show that the radiation heat transfer using N radiation shields is $\frac{1}{N+1}$ times of radiation heat transfer using no shield between two parallel plates when all emissivities are equal to 1. 10

3(c). A surface is shaped like a long equilateral triangle duct. The width of each side is 1 m. The base surface has an emissivity of 0.7 and is maintained at a uniform temperature of 600 K. The heated left side surface closely approximates a blackbody at 1000 K. The right side surface is well insulated. Determine the rate at which heat must be supplied to the base surface externally per unit length of the duct in order to maintain these operating conditions. 12

3(d). What is geometric factor? How to estimate monthly average daily radiation on tilted surface? 08

4(a). What is a solar collector? Classify solar collectors according to the range of temperature of applications. 06

4(b). With the help of a schematic diagram, explain the working principle of a flat plate solar collector. 10

4(c). Briefly discuss about the materials used for different components of: 10

(i) Flat plate collector, and (ii) Concentrating collector.

4(d). "Evacuated tube solar collectors offer better performance than FPCs". Justify the statement. 09

SECTION – B

5(a). Define the followings which are related to solar collector: 08

(i) Critical radiation level, (ii) Collector efficiency factor, (iii) Collector heat removal factor, and (iv) Semi-acceptance angle.

5(b). With the help of a graphical/schematical representation, explain the significance of the followings: 12

(i) Concentration ratio, and (ii) Effective transmittance-absorptance product.

5(c). With the help of necessary figure, classify the concentrating solar collector. 10

5(d). What are the advantages and disadvantages of concentrating collector? 05

6(a). Discuss about all the losses of a flat plate collector. 12

6(b). Explain the construction of a heat pipe evacuated tube collector with necessary diagram. 10

6(c). Why tracking is essential for concentrating solar collectors? Briefly discuss the appropriate types of tracking system for concentrating solar collector. 13

7(a). Explain the working principle of a direct solar water heating system. 10

7(b). Explain the problem associated with integrated solar collector storage system and its possible solution. 08

7(c). Write down the advantages of solar crop-drying. Describe the working principle of solar crop-drying system with schematic diagram. 10

7(d). Briefly explain the solar distillation process. 07

8(a). Even though solar thermal collectors convert the solar radiation into heat, utilizing this heat, explain any one process for the purpose of cooling. 11

8(b). What are the differences between solar pool heating and solar pond heating systems? 06

8(c). Schematically show and discuss the central receiver system to generate electricity. 09

8(d). Briefly explain the basic construction of a PVT crop-drying system. 09

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 1st Term Examination, 2023

EE 3113

(Power System Engineering)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a).	What do you understand by the parametres of over head transmission lines? Why inductance and capacitances are formed in transmission line?	10
1(b).	What are the advantages and disadvantages of dc transmission?	08
1(c).	Derive the expressions for transmission efficiency and volume of conductor mterial with respect to transmission voltage. What are the effects of high transmission voltage on those parameters.	17
2(a).	Determine the inductance of a single phase two wire transmission line.	15
2(b).	What is visual ccritical voltage? What are the advantages of corona.	08
2(c).	Explain the following i) Proximity Effect, ii) Skin Effect, iii) Effect of earth on capacitance.	12
3(a).	Draw the vector diagram of a medium voltage transmission line represented by a T section. Consider that the receiving end voltage V_R is known and the purpose of the vector diagram is to determine the sending end voltage V_S.	12
3(b).	What factors need to be considered for determining the tower height, conductor clearance to ground and sag-tension.	10
3(c).	An overhead transmission line at a river crossing is supported by two towers at heights of 60 m and 90 m above water level, the horizontal distance between the towers being 400 m. If the maximum allowable tension is 2000 kg, find the clearance between the conductor and water at a point mid-way between the towers. Weight of conductor is 1 kg/m.	13
4(a).	What is corona? How can the effect of corona be reduced?	08
4(b).	Why are insulators used with overhead lines? Discuss the desirable properties of insulators.	12
4(c).	Each line of a 3-phase system is suspended by a string of 3 identical insulators of a self-capacitance C farad. The shunt capacitance of connecting metal work of each insulator is 0·2C to earth and 0·1C to line. Calculate the string efficiency of the system if a guard ring increases the capacitance to the line of metal work of the lowest insulator to 0·3 C.	15

SECTION – B

5(a).	What is phase sequence in power system? Explain why it is important to ensure the correct phase sequence when connecting 3- phase motors to the grid.	08
5(b).	What are the causes of faults in power systems? Describe why fault analysis is essential?	06
5(c).	Derive an expression for fault current of single line-to-ground faults by using the concept symmetrical components methods and draw the sequence network connection for the single line-to-ground fault.	13
5(d).	What is the reason of transmission interconnection? What is the impact of cyber security on the smart grid?	08

6(a). Discuss the importance of voltage control in the modern power system. Describe the on-load tap changing transformer method of voltage control. 10

6(b). A load of 10,000 kW at a power factor of 0·8 lagging is supplied by a 3-phase line whose voltage has to be maintained at 33kV at each end. If the line resistance and reactance per phase are 5 Ω and 10 Ω respectively, calculate the capacity of the synchronous condenser to be installed for the purpose. 05

6(c). Explain the significance of the five pins in a five-pin relay, including their respective functions. 10

6(d). What is a sub-station? Name the factors that should be taken care of while designing and erecting a substation. What is the need of a sub-station in the power system? 10

7(a). Explain the concept of arc extinction in a circuit breakers and why it is important for the safe operation of electrical systems? 07

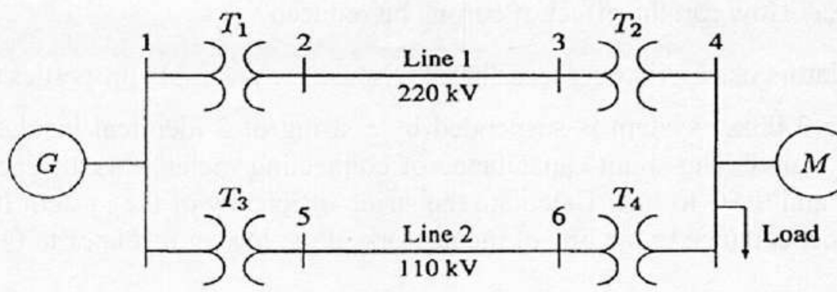
7(b). Why SF_6 CB is preferred? Explain the construction and principle of operation of Sulphur Hexafluoride (SF_6) circuit breakers. 12

7(c). A 50 Hz, 11 kV, 3-phase alternator with earthed neutral has a reactance of 5 ohms per phase and is connected to a bus-bar through a circuit breaker. The distributed capacitance upto circuit breaker between phase and neutral in 0·01 μF. Determine 08
(i) peak re-striking voltage across the contacts of the breaker
(ii) frequency of oscillations
(iii) the average rate of rise of re-striking voltage upto the first peak

7(d). Why do use C.T in the relay circuit? Discuss the different types of bus-bar arrangements. 08

8(a). What is lighting arresters? How does a lighting arrester mitigate the risks associated with lighting strikes? 09

8(b). The one-line diagram of a 3-phase power system is shown in figure. Select a common base of 100 MVA and 22 kV on the generator side. Draw an impedance diagram with all impedances including the load impedance marked in per-unit. The manufacturer's data for each device is given as follow. 14



G:	90 MVA	22 kV	$X = 18\%$
T_1:	50 MVA	22/220 kV	$X = 10\%$
T_2:	40 MVA	220/11kV	$X = 6.0\%$
T_3:	40 MVA	22/110 kV	$X = 6.4\%$
T_4:	40 MVA	110/11kV	$X = 8.0\%$
M:	66.5 MVA	10.45 kV	$X = 18.5\%$

The three phase load at bus 4 absorbs 57MVA, 0.6 power factor lagging at 10.45 kV line 1 and line 2 have reactances of 48.4 and 65.43 Ω, respectively.

8(c). What do you mean by grounding or earthing? Explain it with an example. 07

8(d). Draw the key diagram of a typical 11 kV/400 V indoor sub-station. 05

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2023

ESE 3123

(Thermo-Fluid Devices)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks. Required graph /table will be provided
iii) Assume reasonable data if any missing.

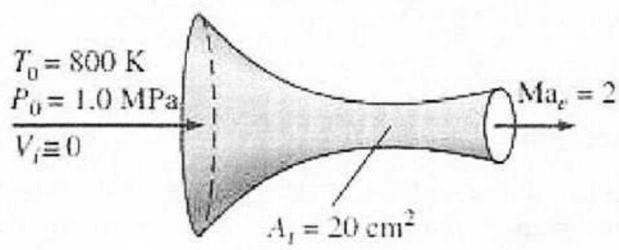
SECTION – A

1(a). Define the following terms: (i) Compressible flow, (ii) Sonic flow and (iii)Shock waves 06

1(b). What is stagnation pressure? Derive an expression for stagnation pressure in compressible flow. 09

1(c). Define Sound. Prove that the velocity of sound in an ideal gas for one dimensional isentropic flow in compressible fluid is given by: $c = \sqrt{\gamma \frac{P}{\rho}}$, where γ is the adiabatic index, P is the pressure, and ρ is the density of the gas. 09

1(d). Air enters a converging–diverging nozzle at 1.0 MPa and 800 K with negligible velocity. The flow is steady, one-dimensional, and isentropic with k = 1.4. For an exit Mach number of Ma = 2 and a throat area of 20 cm^2, determine (i) the throat conditions, (ii) the exit plane conditions, including the exit area, and (iii) the mass flow rate through the nozzle. 11



2(a). Show that the equation of Rayleigh line is 08

$$\frac{P_2}{P_1} = \frac{1 + kMa_1^2}{1 + kMa_2^2}$$

(The symbols have their usual meanings)

2(b). Helium following steadily in a nozzle experience a normal shock at a Mach number of Ma = 2.5. If the pressure and temperature of Helium are 61.64 kPa and 262.15 K, respectively, upstream of the shock, calculate the pressure , temperature , velocity , Mach number and stagnation pressure downstream of the shock. For Helium use k = 1.667 and R = 2.0769 kJ/kg.K 12

2(c). Describe the purpose of using a compressor and explain the working principles of centrifugal and reciprocating compressors. 15

3(a). What are the Affinity Laws? Illustrate and explain the characteristic curve of a centrifugal pump. 08

3(b). Define Pump. Illustrate and describe the working principle of a single-stage centrifugal pump. 09

3(c). Derive an expression for the work done by the impeller of a centrifugal pump on the liquid per second per unit weight of the liquid. 10

3(d). A pump in a current system operates at a flow rate of 80 gpm with a maximum delivered pressure head of 90 ftH_2O. The power demand is 6 hp at an impeller velocity of 750 rpm. After system modification, the total system pressure head increases to 120 ftH_2O. Determine the impeller velocity needed to overcome the higher system backpressure. 08

4(a). Explain cavitation in the context of pump operation and discuss how NPSH is related to cavitation. 08

4(b). Compare and contrast the advantages and disadvantages of plunger pumps and diaphragm pumps in terms of their applications and performance characteristics. 08

4(c). Define the concepts of slip and percentage slip in reciprocating pumps. Using diagrams, describe the operational principle of a reciprocating pump. 10

4(d). A single-acting reciprocating pump operates at 50 rpm and has a stroke length of 40 cm. The piston diameter is 25 cm, and it discharges 0.015 m^3/s of water. Determine: (i) Theoretical discharge of the pump, (ii) Slip and percentage slip of the pump, (iii) Coefficient of discharge and (iv) Power required to run the pump. 09

SECTION – B

5(a). State the Buckingham's π theorem. 04

5(b). What is orifice meter? Derive a mathematical expression to calculate flow rate using orifice meter. 15

5(c). Describe the relative accuracy of venturimeter nozzle meter and orifice meter. 06

5(d). A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm respectively s used to measure the flow of water. The reading of differential manometer connected to the inlet and throat is 20 cm of mercury. Determine the flow rate. Take $C_d = 0.98$. 10

6(a). Write short notes on the following term: 06

(i) Irrational flow region, (ii) hydrodynamic entrance regions and (iii) hydrodynamically fully developed region.

6(b). With appropriate force balance show that for fully developed laminar flow in a circular pipe the velocity profile is given by $\mu(r) = 2V_{avg}(1 - \frac{r^2}{R^2})$ 15

(The symbols have their usual meanings)

6(c). Differentiate between major and minor losses in pipe-flow. 04

6(d) Water at 10°C ($\rho = 999.7\,kg/m^3$ and $\mu = 1.307 \times 10^{-3}\,kg/m.s$) is flowing steadily in a 0.2 cm diameter, 15 m long pipe at an average velocity of 1.2 m/s. Determine (i) the pressure drop, (ii) the head loss and (iii) the pumping power required to overcome this pressure drop. 10

7(a). What is heat exchange? Write down its applications. 05

7(b). Show that the heat transfer rate in a counter-flow heat exchanger can be expressed by: 10

$\dot{Q} = UAs\Delta T_{lm}$ (the symbols have their usual meanings)

7(c) A counter flow, concentric tube heat exchanger is designed to heat water from 20 to 80°C using hot oil, which is supplied to the annulus at 160°C and discharged at 140°C. The thin walled inner tube has a diameter of $D_i = 20$ mm, and the overall heat transfer coefficient is 500 W/ m^2. K. The design condition calls for a total heat transfer rate of 3000 W. (i) What is the length of the heat exchanger?, (ii) After 3 years of operation, performance is degraded by flowing on the water side of the heat exchanger, and the water outlet temperature is only 65°C for the same fluid rates and inlet temperature. What are the corresponding values of the heat transfer rate, the outlet temperature of the oil, the overall heat transfer coefficient and the water side fouling factor? 20

8(a). Draw the temperature profiles for counter flow and parallel flow heat exchangers. Explain why a counter flow heat exchanger is more effective in case of heat transfer? 08

8(b). Show that the general expression of effectiveness for parallel flow heat exchanger is 15

$$\varepsilon = \frac{1 - e^{-NTU\left(1 + \frac{C_{min}}{C_{max}}\right)}}{1 + \frac{C_{min}}{C_{max}}}$$

(The symbol have their usual meaning)

8(c). In a daily operation, milk at a flow rate 250 liter/hour and cow body temperature of 38.6°C must be chilled to a safe-to-store temperature of 13°C. Ground water at 10 °C is available at a flow rate of 0.72 m^3/h. The density and specific hat of milk are 1030 kg/m^3 and 3860 J/kg. K, respectively. (i) Determine the UA product of a counter flow heat exchanger required for the chilling process. Determine the length of the heat exchanger if the inner pipe has a 50 mm diameter and the overall heat transfer efficient is U =1000 $W/m^2.K$, (ii) Determine the outlet temperature of the water. 12

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2023

ESE 3105

(Heat and Mass Transfer)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks. Required graph /table will be provided
iii) Assume reasonable data if any missing.

SECTION – A

1(a). What are the basic modes of heat transfer? Describe the modes by mentioning their governing equations. 08

1(b). Derive the general heat conduction equation for a rectangular co-ordinate system. 14

1(c). The surface of a plane wall of thickness L are maintained at temperature t_1 and t_2. The thermal conductivity of the wall materials varies according to the relations: 13

$$k = k_0 t + 0.5 k_0 t^2$$

(i) Derive an expression to find the steady state condition through the wall.
(ii) Find the temperature at which the mean thermal conductivity is evaluated in order to get the same heat flow by its substitution in the simplified Fourier's equation

2(a). What is meant by overall heat transfer coefficient? Explain how it can be measured by deducing the expression for overall heat transfer coefficient. 12

2(b). Write down the significance of critical thickness of insulation. Also, Derive the expression for critical thickness of a cylindrical object. 12

2(c). Consider a 3-m high, 6-m wide and 0.25m thick brick wall whose thermal conductivity is $K = 0.8\ \mathrm{W/m.k}$. On a certain day the temperature at the inner and outer surfaces of the wall measured to be 14°C and5 °C, respectively. Determine the rate of heart loss through the wall on that day. 11

3(a). What is meant by effectiveness and efficiency of a fin? "An increase in fin effectiveness can be obtained by increasing the length of the fin but it decreases the efficiency of the fin"- Justify the statement. 12

3(b). A motor body is 400 mm in outside diameter and 250 mm long. Its surface temperature should not exceed 62°C when dissipating 350 W. Longitudinal fins of 18 mm thickness and 38 mm height are proposed. The convective coefficient is 42 $\mathrm{W/m^2 °C}$. Now, determine the number of fins required. Atmospheric temperature is 35°C and thermal conductivity is 40 W/m°C. 08

3(c). A long 35cm diameter cylindrical shaft made of stainless steel 304 ($k = 14.9\ \mathrm{W/m°C}, \rho = 7900\ \mathrm{kg/m^3}, c_p = 477\ \mathrm{J/kg°C}$ and $\alpha = 3.95 \times 10^{-6}\ \mathrm{m^2/s}$) comes out of an oven at a uniform temperature of 400°C. The shaft is then allowed to cool slowly in a chamber at 150 °C with an average convection heat transfer coefficient of $h = 60\ \mathrm{W/m^2 °C}$. Determine the temperature at the center of the shaft 20 min after the start of cooling process. Also, determine the rate of heat transfer per unit length of the shaft during this time period. 15

4(a). Differentiate between film and dropwise condensation. Also describe the film condensation phenomena for a vertical plate. 10

4(b). Describe the pool boiling phenomena with necessary diagram by mentioning each stage of the diagram. 10

4(c). A large condenser has several column of the tube with 33 tubes in each column. The outer diagram of the tube is 1.5 cm. The condenser operates at a pressure of 12.35 kPa and the outer surface of the tubes are maintained at 20°C. Determine i) Average heat transfer coefficient, ii) the rate of condensation of steam per unit length of column 15

$$h = 0.729\left[\frac{g\rho_l(\rho_l - \rho_v)h^*_{fg}k_l^3}{\mu(T_{sat}\text{-}T_s)D}\right]^{1/4}$$

SECTION – B

5(a). With the help of necessary sketch write down the physical mechanism of convection heat transfer. 10

5(b). Differentiate between hydrodynamic and thermal boundary layer. "The thickness of the thermal boundary layer increases in the flow direction, but heat transfer coefficient decreases with increasing the distance "- Justify the statement. 12

5(c). During the flow of air at $T_\infty = 20°C$ over plate surface maintained at a constant temperature of $T_s = 160°C$. The dimensionless temperature profile within the air layer over the plate is determined to be 13

$$\frac{T(y) - T_\infty}{(T_s - T_\infty)} = e^{-ay}$$

Where $a = 3200\text{m}^{-1}$ and y is the vertical distance measured from the plate surface in m. Determine the heat flux on the plate surface and the convection heat transfer coefficient.

6(a). Write down the significance of following dimensionless number: 08

(i) Prandtl Number, (ii) Rayleigh Number, (iii) Nusselt Number

6(b). Consider a $0.6m \times 0.6m$ thin square plate in a room at 30°C. One side of the plate is maintained at a temperature of 90°C while the other side is insulated. Determine the rate of heat transfer from the plate by natural convection if the plate is (i) vertical, (ii) horizontal with hot surface facing up, (iii) horizontal with hot surface facing down 15

$$[N_u = 0.59Ra_L^{1/4}, N_u = 0.27Ra_L^{1/4}, N_u = 0.15Ra_L^{1/4}, N_u = 0.1Ra_L^{1/3}, N_u = 0.54Ra_L^{1/4}]$$

$$N_u = \left[0.64 \frac{0.387Ra^{1/6}}{\left[1 + \left(0.559/_{Pr}\right)^{9/16}\right]^{8/27}}\right]^2$$

6(c). Derive the energy equation for the laminar boundary layer on a flat plate. 12

7(a). What are the analogy between heat transfer and mass transfer? Give example for i) liquid -to -gas, ii) solid -to - gas mass transfer. 10

7(b). In an industrial facility air is to be preheated before entering a furnace by geothermal water at 120° C and flowing through the tubes of a tube bank located in a duct. Air enters the duct at 20 °C and 1 atm with a mean velocity of 4.5 m/s, and flows over the tubes in natural direction. The outer diameter of the tubes is 1.5 cm and the tubes are arranged in-line with longitudinal and transverse pitches of $S_L = S_T = 5$ cm. There are 6 routs in the flow direction with 10 tubes in each row. Determine the rate of heat transfer per unit length of the tubes, and the pressure drop across the tube bank. 25

8(a). State's the Fick's law of diffusions. Write down the difference between absorption and adsorption. 08

8(b). A deep narrow cylindrical vessel which is open the top contains some toluene at the bottom. The air within the vessel is considered motionless, but there is sufficient air current at the top surfaces of the vessel that any toluene vapor arriving at the top surfaces is immediately removed to ensure zero toluene concentration at the surface. The entire system is at atmospheric pressure and 18.7°C. Under these conditions $D = 0.826 \times 10^{-5}\ m^2/s$ and the saturated vapor pressure of toluene at the liquid surface in the vessel is 0.026 atm. Determine the rate of evaporation of toluene into the air per unit area of the liquid surface if the distance between the liquid toluene surface and the top of the vessel is 1.524 m. 17

8(c). "For thermally fully developed flow, the axial temperature gradient is independent of the radial direction"– Justify the statement. 10

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